

Research Proposal: Detecting Insider Threats through LLM-Based Embeddings and Anomaly Detection

Introduction

Insider threats represent a critical security challenge for modern organisations. Because insiders act within authorised boundaries, their malicious behaviour blends into a large volume of legitimate activity, and genuine incidents are exceedingly rare. Email is a primary channel through which such activity occurs (data exfiltration, unauthorised disclosure, etc.) making it a natural focus for detection.

This research proposes a framework that combines Large Language Model (LLM) embeddings with Anomaly Detection methods to identify suspicious email communications. Rather than prescribing a single algorithm, the project treats detection as a rare event problem and invites a progression of modelling approaches, from simple statistical baselines to more advanced Machine Learning and Deep Learning techniques.

The key idea is to represent each user's email behaviour through both metadata and the semantic content of their messages, and then to detect deviations from normal patterns.

Research Objectives

- 1. Convert Raw Emails to Meaningful Representations:** Extract semantic features from unstructured email text using pre-trained LLMs to generate high-dimensional vector representations (embeddings) that capture characteristics such as tone, topic, and deviation from a user's normal communication patterns. These can be combined with behavioural metadata (volume, recipients, size, attachments, timing).
- 2. Build an Anomaly-Detection Framework:** Develop a detection framework that flags emails or users deviating from established norms. The modelling approach is deliberately left open-ended, beginning with simple, interpretable methods and progressing toward more advanced Machine Learning and Deep Learning techniques, so that students can explore and compare trade-offs.
- 3. Experimenting with the CERT Insider Threat dataset:** Test the system on the CERT Insider Threat Dataset and measure how effectively threats are detected under severe class imbalance while respecting realistic analyst workload constraints.

Methodology

The methodology comprises three stages, each stated as a recommended direction rather than a fixed procedure.

Stage 1 - Feature Engineering

Emails are aggregated per user over a defined temporal window to characterise behaviour. Two complementary representations are recommended: quantitative metadata features (volume, recipients, size, attachments, timing) and semantic features obtained from pretrained language model embeddings of message content. Features capturing deviation from each user's historical baseline are expected to be particularly informative (dimensionality reduction may be applied to improve efficiency).

Stage 2 - Modelling

The modelling strategy is deliberately open-ended. Simple, interpretable baselines are recommended to establish reference performance, followed by classical anomaly detection and supervised methods, and subsequently embedding driven or deep learning approaches.

Stage 3 - Training and Evaluation.

Given the sequential structure of the data, temporally ordered partitioning is recommended to prevent leakage, with normalisation estimated on training periods only. Evaluation should reflect operational conditions, calibrating detection to a realistic analyst workload.

Dataset

The project will utilize the **CERT Insider Threat Dataset (v6.2)**, a synthetic dataset developed by the CERT Division (Software Engineering Institute). It provides realistic **background activity** for normal users alongside injected **malicious actor scenarios**, avoiding the privacy and labelling problems of real data.

The project uses release **r6.2**, the most recent and extensive version, containing **4,000 users**. Students can download the dataset in compressed form and extract the data folder to access the log files. An accompanying **answers** file lists the malicious scenarios and the identifiers of the users involved, providing ground truth labels for evaluation.

CERT dataset has two main sources which are:

- User activities which includes; e-mail, http, logon, file
- Organisation structure and user profile information (LDAP)

Insider threat detection generally works best when behaviour is aggregated per user over a time window, and combining email content with the other activity signals (web, logon etc.) would produce the strongest anomaly-detection models. However, to keep the project manageable for students, its scope is deliberately limited: we recommend using the email logs (email.csv) for anomaly detection, enriched with the LDAP organisational and user-profile data for context.

Key properties:

- Content is written as complete sentences, so it suits both language-model embeddings and classical text pre-processing (tokenisation, stop-word removal, stemming).
- Email traffic follows an underlying organisational graph: most sender–recipient pairs are genuine, with a few random edges as noise, and recipients may be internal (DTAA) or external (non-DTAA). Off-graph or external recipients, and unusual volume, size, or attachment counts, act as natural anomaly signals.
- Malicious insiders are extremely rare, which frames the task as rare-event anomaly detection.

Downloading and Extracting the Dataset

The dataset files can be downloaded from the CERT Insider Threat Test Dataset page on [Insider Threat Test Dataset - Carnegie Mellon University](#).

Due to the size of the dataset, it is often easier to extract the compressed archive from the terminal rather than through a GUI. For example, the r6.2 archive can be unzipped into a chosen folder with:

```
tar -xvf ~/Downloads/r6.2.tar.bz2 -C ~/Downloads
```